

Assignment 7 (ELEC 341 L7_RouthHurwitzEx)

Problem 1:

Tell how many roots of the following polynomial are in the right half-plane, in the left half-plane, and on the $j\omega$ -axis:

$$P(s) = s^5 + 6s^3 + 5s^2 + 8s + 20 = s^5 + \varepsilon s^4 + 6s^3 + 5s^2 + 8s + 20.$$

Use Routh-Hurwitz criterion.

s^5	1	6	8
s^4	ε	5	20
s^3	$\frac{6\varepsilon - 5}{\varepsilon}$	$\frac{8\varepsilon - 20}{\varepsilon}$	0
s^2	$G_0(\varepsilon)$	20	0
s^1	$G_1(\varepsilon)$	0	0
s^0	20	0	0

just
plug # in
much easier.

$$G_0(\varepsilon) = \frac{\left(\frac{30\varepsilon - 25}{\varepsilon}\right) - (8\varepsilon - 20)}{\frac{6\varepsilon - 5}{\varepsilon}}$$

$$G_0(\varepsilon) = \frac{30\varepsilon - 25 - 8\varepsilon^2 + 20\varepsilon}{6\varepsilon - 5}$$

$$G_0(\varepsilon) = \frac{-8\varepsilon^2 + 50\varepsilon - 25}{6\varepsilon - 5}$$

$$\lim_{\varepsilon \rightarrow 0} G_0(\varepsilon) = \lim_{\varepsilon \rightarrow 0} \frac{-8\varepsilon^2 + 50\varepsilon - 25}{6\varepsilon - 5} = +5$$

$$G_1(\varepsilon) = \frac{G_0(\varepsilon) \cdot \frac{8\varepsilon-20}{\varepsilon} - 20 \left(\frac{6\varepsilon-5}{\varepsilon} \right)}{G_0(\varepsilon)}$$

$$G_1(\varepsilon) = \left[\frac{-8\varepsilon^2 + 50\varepsilon - 25}{6\varepsilon - 5} \cdot \frac{8\varepsilon - 20}{\varepsilon} - 20 \left(\frac{6\varepsilon - 5}{\varepsilon} \right) \right] \times \frac{6\varepsilon - 5}{-8\varepsilon^2 + 50\varepsilon - 25}$$

$$= \frac{(-8\varepsilon^2 + 50\varepsilon - 25)(8\varepsilon - 20) - 20(6\varepsilon - 5)^2}{\varepsilon(6\varepsilon - 5)} \times \frac{6\varepsilon - 5}{-8\varepsilon^2 + 50\varepsilon - 25}$$

$$= \frac{(-8\varepsilon^2 + 50\varepsilon - 25)(8\varepsilon - 20) - 20(6\varepsilon - 5)^2}{\varepsilon(-8\varepsilon^2 + 50\varepsilon - 25)}$$

$$720\varepsilon^2 + 1200\varepsilon - 500$$

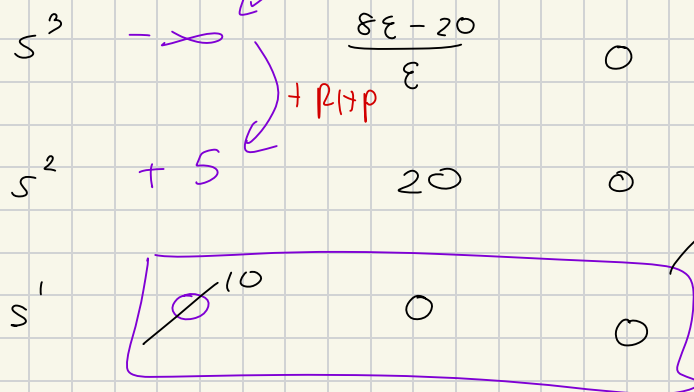
$$G_1(\varepsilon) = \frac{(-64\varepsilon^3 + 160\varepsilon^2 + 400\varepsilon^2 - 1000\varepsilon - 200\varepsilon + 500) - 20(36\varepsilon^2 - 60\varepsilon + 25)}{\varepsilon(-8\varepsilon^2 + 50\varepsilon - 25)}$$

$$= \frac{-64\varepsilon^3 - 160\varepsilon^2}{\varepsilon(-8\varepsilon^2 + 50\varepsilon - 25)}$$

$$G_1(\varepsilon) = \frac{64\varepsilon^2 + 160\varepsilon}{8\varepsilon^2 - 50\varepsilon + 25}$$

$$\lim_{\varepsilon \rightarrow 0} G_1(\varepsilon) = \lim_{\varepsilon \rightarrow 0} \frac{64\varepsilon^2 + 160\varepsilon}{8\varepsilon^2 - 50\varepsilon + 25} = 0$$

$$\Rightarrow \begin{array}{ccccc} s^5 & 1 & & 6 & 8 \\ s^4 & \varepsilon & & 5 & 20 \\ s^3 & -\infty & \xrightarrow{+P+P} & \frac{8\varepsilon-20}{\varepsilon} & 0 \\ s^2 & +5 & \xrightarrow{+P+P} & 20 & 0 \\ s^1 & & & 0 & 0 \\ s^0 & 20 & & 0 & 0 \end{array} \quad \varepsilon > 0$$



$$P(s) = 5s^2 + 20 = 0$$

$$P'(s) = 10s$$

$$5s^2 + 20 = 0$$

$$s^2 = -4$$

$$s = \pm 2i$$

2 poles on RHP
2 poles on jw axis
1 pole on LHP

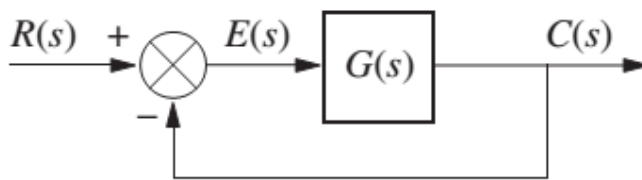
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Problem 2:

In the system of Figure, let

$$G(s) = \frac{K(s+2)}{s(s-1)(s+3)}$$

Find the range of K for closed-loop stability.



FIGURE

$$\frac{C(s)}{R(s)} = \frac{K(s+2)}{s(s-1)(s+3)} \cdot \frac{1}{1 + \frac{K(s+2)}{s(s-1)(s+3)}}$$

Use Routh-Hurwitz criterion.

$$(s-1)(s+3) = s^2 + 2s - 3$$

$$1 + \frac{K(s+2)}{s(s-1)(s+3)} = 0$$

$$\Rightarrow s^3 + 2s^2 - 3s + Ks + 2K = 0$$

$$\Rightarrow s^3 + 2s^2 + (K-3)s + 2K = 0$$

$$\begin{array}{cccc} s^3 & 1 & K-3 & 0 \end{array}$$

$$\begin{array}{cccc} s^2 & 2 & 2K & 0 \end{array}$$

$$\begin{array}{cccc} s^1 & -3 & 0 & \end{array}$$

$$\begin{array}{cccc} s^0 & 2K & \Rightarrow & 2K > 0 \\ & & & K > 0 \end{array}$$

$$\frac{2K - 6 - 2K}{2} = -3$$

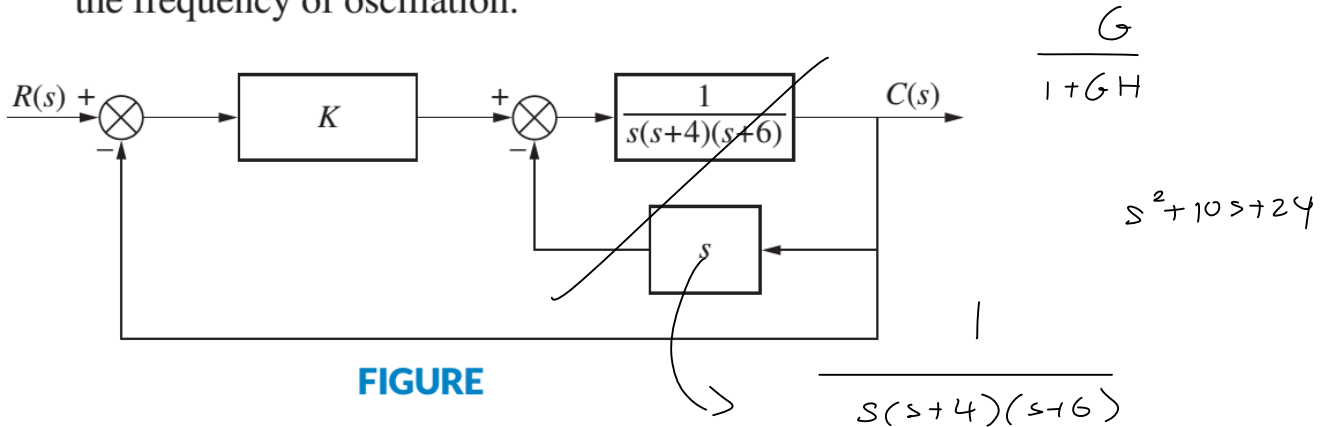
1) but due to sign change

2) no value of K will make the system stable.

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Problem 3:

For the system shown in Figure, find the value of gain, K , that will make the system oscillate. Also, find the frequency of oscillation.



Use Routh-Hurwitz criterion.

$$\frac{C(s)}{R(s)} = \frac{K}{s^3 + 10s^2 + 24s + K}$$

$$= \frac{1}{s^3 + 10s^2 + 24s + K}$$

$$\Rightarrow s^3 + 10s^2 + 24s + K = 0$$

$$s^3 \quad 1 \quad 24$$

$$s^2 \quad 10 \quad K$$

$$s^1 \quad \frac{250-K}{10} \quad 0$$

$$s^0 \quad K$$

$$\frac{250-K}{10} > 0$$

$$250 > K$$

$$K < 250$$

$$0 < K < 250$$

at marginal stability $K = 250$ or K value for oscillation.

$$\Rightarrow s^3 \quad 1$$

$$s^2 \quad 10 \quad 250$$

$$s^1 \quad 0 \quad 0$$

$$s^0 \quad 250$$

$$\Rightarrow P(s) = 10s^2 + 250 = s^2 = -25$$

$$s = \pm 5i$$

$$\omega = 5 \text{ rad/s}$$

$\hookrightarrow$ frequency of oscillation.